# MPG No. 8.7.126 $m'mm$ (mm'm setting) [ Type III, orthorhombic ] [T tensor]

* Rank 0 tensor. * Rank 1 tensor.

$$\begin{bmatrix} 0 & T_y & 0 \end{bmatrix}$$

$$T_y = \mathbb{T}_y^{(1)}$$

* Rank 2 tensor (s). * Rank 2 tensor (a). * Rank 3 tensor (ss).

$$\begin{bmatrix} 0 & T_{xxy} & 0 \\ 0 & T_{yyy} & 0 \\ 0 & T_{zzy} & 0 \\ 0 & 0 & T_{zzy} \\ 0 & 0 & 0 \\ T_{xxy} & 0 & 0 \end{bmatrix}$$

$$T_{xxy} = \mathbb{T}_y^{(1)} - \mathbb{T}_y^{\alpha(1)} - \mathbb{T}_y^{\beta(1)}$$
$$T_{yyy} = 3\mathbb{T}_y^{(1)} + 2\mathbb{T}_y^{\alpha(1)}$$
$$T_{zzy} = \mathbb{T}_y^{(1)} - \mathbb{T}_y^{\alpha(1)} + \mathbb{T}_y^{\beta(1)}$$

* Rank 3 tensor (sa).

$$\begin{bmatrix} 0 & T_{xxy} & 0 \\ 0 & 0 & 0 \\ 0 & T_{zzy} & 0 \\ 0 & 0 & -\frac{T_{zzy}}{2} \\ 0 & 0 & 0 \\ -\frac{T_{xxy}}{2} & 0 & 0 \end{bmatrix}$$

$$T_{xxy} = -2\mathbb{M}_{xz}^{(1)} + 2\mathbb{T}_y^{(2)}$$
$$T_{zzy} = 2\mathbb{M}_{xz}^{(1)} + 2\mathbb{T}_y^{(2)}$$
$$\frac{T_{zzy}}{2} = \mathbb{M}_{xz}^{(1)} + \mathbb{T}_y^{(2)}$$
$$\frac{T_{xxy}}{2} = -\mathbb{M}_{xz}^{(1)} + \mathbb{T}_y^{(2)}$$

* Rank 3 tensor (a).

$$\begin{bmatrix} 0 & 0 & T_{yzz} \\ 0 & 0 & 0 \\ T_{xyx} & 0 & 0 \end{bmatrix}$$

$$T_{yzz} = \mathbb{M}_{xz}^{(2)} - \mathbb{T}_y^{(3)}$$
$$T_{xyx} = \mathbb{M}_{xz}^{(2)} + \mathbb{T}_y^{(3)}$$

* Rank 4 tensor (sss). * Rank 4 tensor (ssa). * Rank 4 tensor (aas). * Rank 4 tensor (aaa). * Rank 4 tensor (sa). * Rank 4 tensor (as). * Rank 4 tensor (s). * Rank 4 tensor (a). * Rank 4 tensor (t).